VidVaani

Automated, low-cost dubbing of English technical lectures into Hindi

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The problem: technical education is locked in English

- India's best engineering lectures — NPTEL alone hosts **50,000+ hours** — are overwhelmingly in English.
- A large fraction of our students think and learn in Hindi and other Indian languages; a fast English lecture is a barrier, not a resource.
- **NEP 2020** explicitly calls for higher education content in Indian languages.
- Manual dubbing is slow and expensive: NPTEL's human-verified translation effort has taken years to cover a subset of courses in 11 languages.

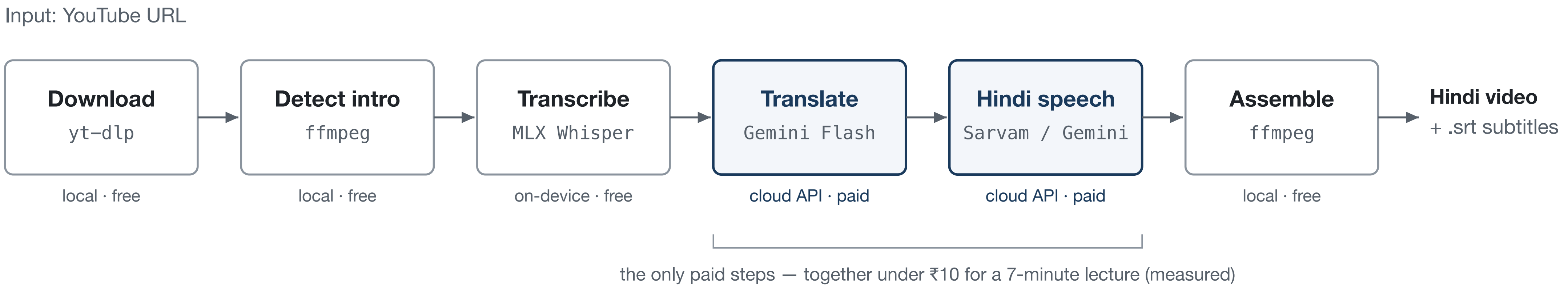
> **Goal:** given any lecture video, produce a natural-sounding Hindi version in minutes, for rupees — with the technical vocabulary students actually use.

NPTEL translation initiative: nptel.ac.in/translation · Bhashini (MeitY) has dubbed ~200 NPTEL/SWAYAM courses in 8 languages via human-in-the-loop workflows.

Existing options do not fit the lecture use case

Option	Cost	Why it falls short
YouTube auto-dubbing	Free	Only the **channel owner** can enable it; dubs are uneditable; no control over voice or terminology; widely criticised as robotic
Commercial dubbing SaaS (ElevenLabs, Rask, HeyGen, Dubverse, Murf)	**₹3,000–11,000 per lecture**	Priced for short marketing clips; monthly minute caps; each extra language billed again
Azure Video Translation	≈ ₹2,000 per lecture	Cheapest managed option, but limited voice choice and no terminology control

VidVaani: a six-stage open pipeline

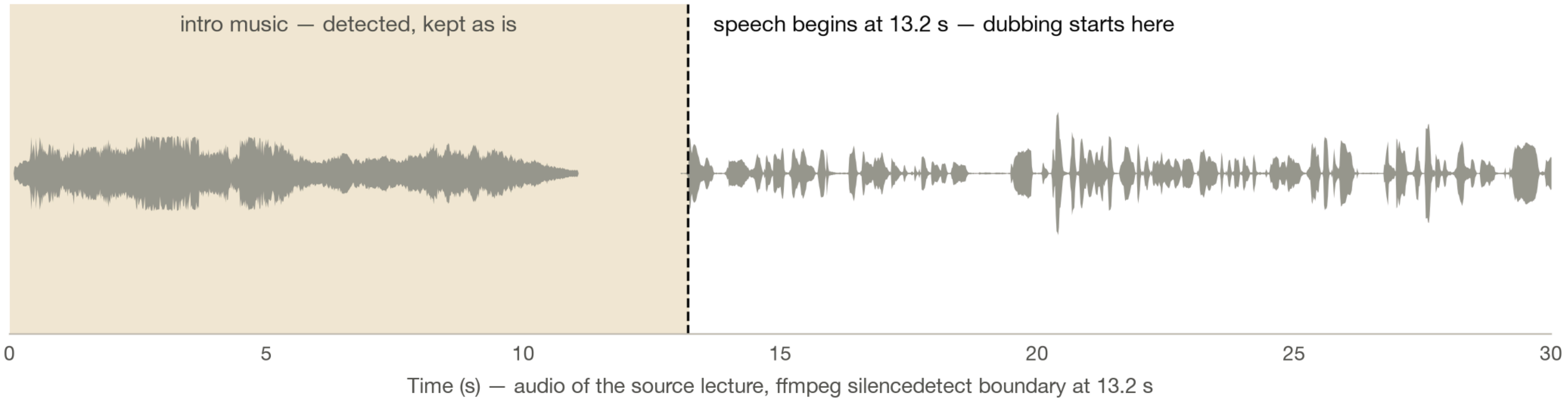


- Single command: `vidvaani dub <URL> --full -b sarvam -v abhilash`
- Transcription runs **on-device** (MLX Whisper on Apple Silicon) — audio never leaves the machine until translation.
- Every intermediate artifact is cached: re-dubbing with a new voice skips download, transcription, and translation.

The next five slides walk **one real lecture** — NPTEL CS7015 Lecture 1.1 (Prof. Mitesh

Steps 1–2 — Download, and find where speech starts

`yt-dlp` fetches any YouTube URL or local file — no allowlist, no channel ownership needed. Then `ffmpeg silencedetect` locates the boundary between intro music and speech:



The music is **not dubbed over** — it is kept exactly as published, and dubbing begins on the first spoken word.

Step 3 — Transcribe on-device

MLX Whisper (`distil-large-v3`) runs locally on Apple Silicon — the audio never leaves the machine. Output: time-stamped English segments, grouped to ≤ 15 s:

Time	Whisper output (verbatim)
18.6 – 29.9 s	"In today's lecture is going to be a bit non-technical. We are not going to cover any technical concepts..."
29.9 – 41.1 s	"So, we hear the terms artificial neural networks, artificial neurons quite often these days..."
41.1 – 56.0 s	"And this history contains several spans across several fields, not just computer science..."

Transcribing the full 7-minute lecture took **17–20 s** (measured) — and it is free.

Rows taken verbatim from the pipeline's transcript file (4TG5a_vNKSo_transcript_en.json)

Step 4 — Translate for the clock, and the classroom

Constraints in the translation prompt: *speakable in the same duration; conversational (not literary) Hindi; technical terms stay in English.*

	Original (English)	VidVaani (Hindi)
13.2 s	"Hello everyone, welcome to lecture 1 of **CS7015** which is the course on **deep learning**."	"सभी को नमस्कार, **CS7015** के लेक्चर एक में आपका स्वागत है, जो **डीप लर्निंग** का कोर्स है।"
29.9 s	"...we hear the terms **artificial neural networks**, artificial neurons quite often these days."	"...आजकल हम आर्टिफिशियल न्यूरल नेटवर्क्स, आर्टिफिशियल न्यूरॉन्स शब्द अक्सर सुनते हैं।"

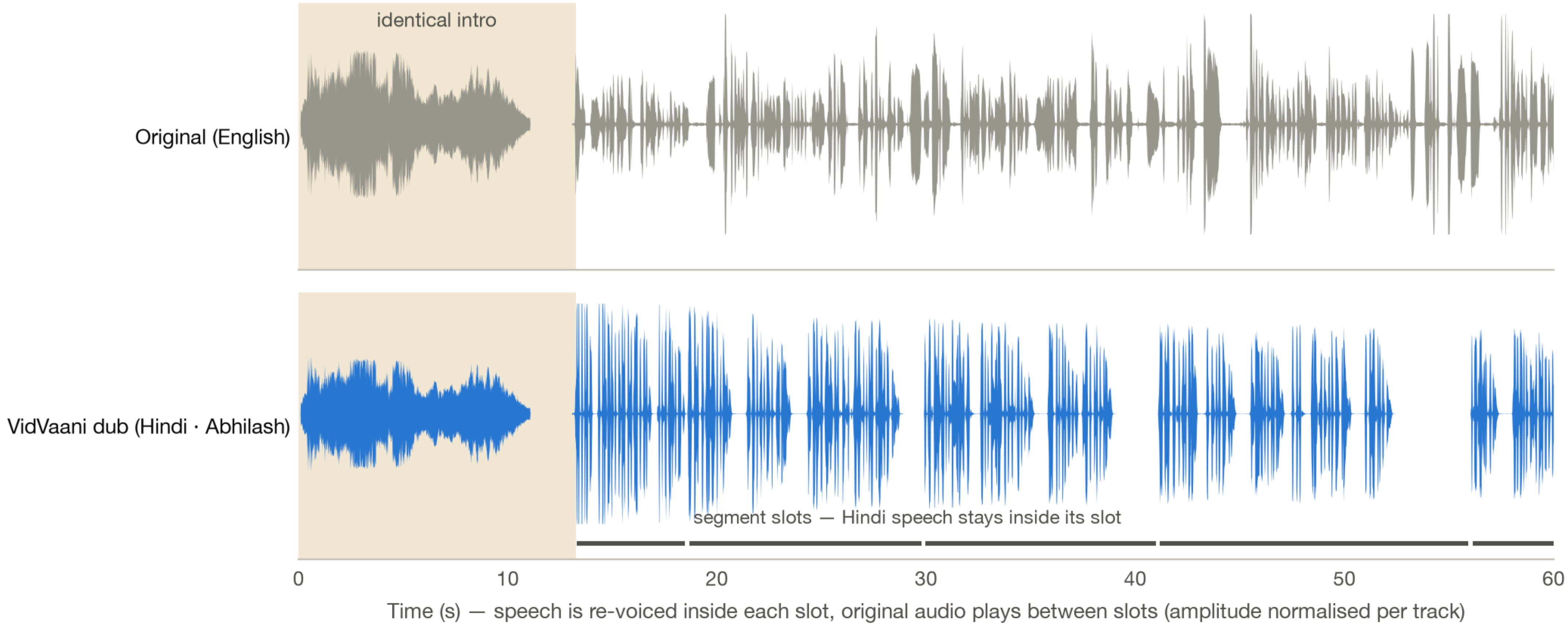
Students say *gradient* and *neural network* — a dub that renders them as प्रवणता and तंत्रिका जाल (as fully-literal systems do) sounds foreign to the very audience it serves.

Step 5 — Synthesize, and fit the clock

Hindi runs ~15–35% longer than English. Per segment, following the automatic-dubbing literature:

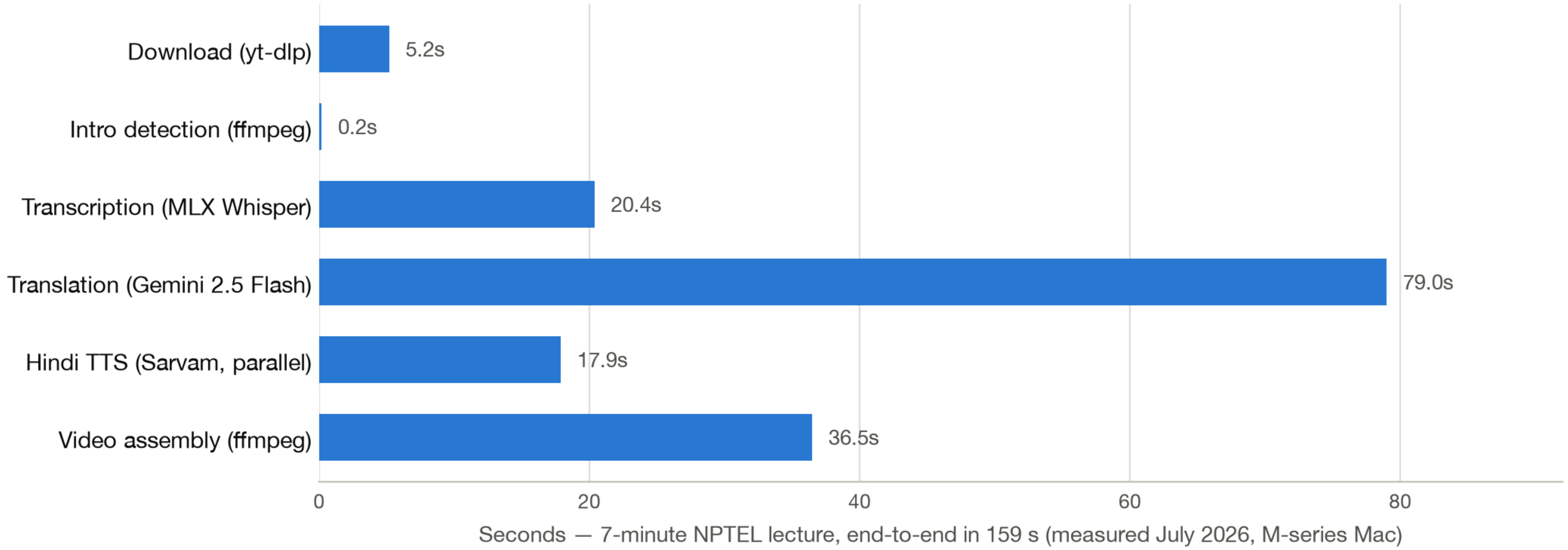
1. The translator gets a **word budget** per slot (2.4 words/s, calibrated against the TTS voice's measured 2.8 words/s delivery) — fixing length at translation time beats stretching audio afterwards.
2. Synthesize at natural pace, trim trailing silence, measure with `ffprobe`.
3. If it fits (±5%) — done. Otherwise speed-adjust with an **asymmetric clamp (0.95×–1.35×)**: listeners tolerate faster speech far better than slowed speech.
4. Still too long? The clip **spills into the trailing pause** rather than being cut — professional dubs sacrifice timing, never content.
5. During pauses, the **original soundtrack plays** — atmosphere and intro music survive the dub.

Step 6 — Reassemble: the result, seen in the signal



The intro is bit-identical to the source; every Hindi utterance starts on its original segment boundary; between utterances the original soundtrack plays. Plus a Hindi `.srt`, and

Measured performance — faster than real time



A **7-minute lecture dubs in 2½–4 minutes** end-to-end on a laptop — the bottleneck is the translation API, not synthesis. Re-runs with a different voice take under a minute (cached transcript + translation).

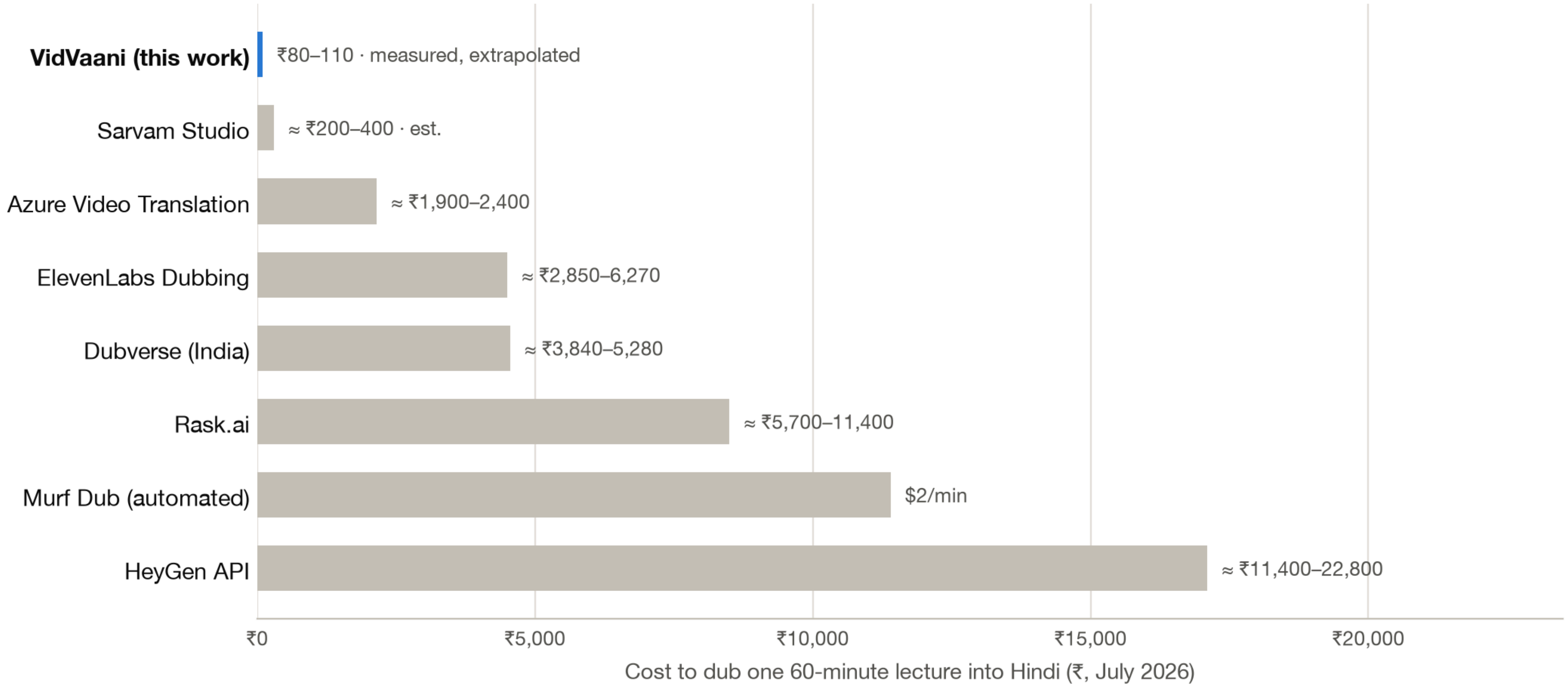
Measured cost — rupees, not lakhs

Measured on the same 7-minute lecture (5,307 Hindi characters):

	Translation	Hindi TTS	Total (7 min)	Extrapolated, 1 hour
Sarvam `bulbul:v2`	₹1.4	₹8.0	**₹9.4** ($0.10)	≈ ₹80
Gemini 2.5 Flash TTS	₹1.4	₹9.6	**₹11.0** ($0.12)	≈ ₹95
Edge TTS (free tier)	₹1.4	₹0	**₹1.4**	≈ ₹12

- A full **40-lecture course ≈ ₹3,200–4,400** — less than one SaaS-dubbed lecture.
- Costs measured by the pipeline itself from API-reported token counts, at official July 2026 prices.
- Stable across runs: after calibrating the translation word budget, the same lecture re

Where this sits in the market



Midpoints of published prices, single target language, July 2026. YouTube auto-dubbing is

Why not just use the platforms?

YouTube auto-dubbing

- Now rolled out to all channels — **but only the uploader** can enable it.
- A university cannot dub NPTEL or MIT content it does not own.
- Dubs cannot be edited — a mistranslated technical term can only be unpublished, not fixed.

Sarvam Studio & SaaS dubbers

- Produce formal, fully-translated Hindi (क्षेत्रफल for "area") — jarring for STEM learners used to Hinglish.
- No control over segment-level phrasing or terminology.
- VidVaani uses the **same Sarvam voices** with full editorial control, at API prices.

What a run looks like

```
$ vidvaani dub "https://youtube.com/watch?v=4TC5s_xNKSs" --full -b sarvam -v anushka

  Downloaded: Deep Learning(CS7015): Lec 1.1 Biological Neuron
  Transcribed: 33 segments          Generated subtitles: 4TC5s_xNKSs_hindi.srt
  TTS: 33/33 segments (parallel)    Created: 4TC5s_xNKSs_hindi_anushka.mp4

            Timing Breakdown                          API Costs
  Download            5.2s    3.3%        Translation   9,542 tokens   $0.015
  Transcription      20.4s   12.8%        TTS (Sarvam)  5,307 chars    Rs 7.96
  Translation        79.0s   49.6%        Total                        Rs 9.4
  TTS Generation     17.9s   11.2%
  Video Assembly     36.5s   22.9%
  Total             159.2s    100%
```

Every run reports its own timing and cost — the numbers on the previous slides are this output, not estimates.

Demonstrations

nipunbatra.github.io/vidvaani — plays in any browser:

1. **58-second calculus clip, eight ways** — original, Sarvam and Gemini voices (same translation, so the comparison is purely the voice), plus Sarvam's own Dashboard output.
2. **Full 7-minute NPTEL lecture** — original beside Hindi dub; intro music preserved; five voices; burned-in subtitles.

What to listen for:

- Technical terms stay in English (*deep learning*, *CS7015*).
- Pauses land where the professor pauses.
- Native Indian prosody (Sarvam) vs. slight Western accent (Gemini).

Honest limitations

- **No human review loop yet** — NPTEL's own effort treats faculty verification as essential for pedagogy; ₹80/lecture buys the draft, not the sign-off.
- **Timing fit is still partly post-hoc** — word budgets plus a 0.95–1.35× speed clamp handle most segments, but very dense passages can still sound hurried (see next slide).
- Gemini TTS models are **preview** — pricing and behaviour may change; Edge TTS is an unofficial free endpoint, fine for prototyping only.
- No lip-sync. Voice cloning of the original lecturer now works locally (next slide) and stays gated on **consent** — the one published cloned demo is the author's own voice on his own lecture.
- Hindi first; Sarvam voices cover 11 Indian languages, so extension is natural.

Towards a fully local pipeline

Every cloud stage now has a working local, open-weights replacement — measured on the 58 s demo clip, Apple Silicon, July 2026:

Stage	Cloud (current)	Local (experimental)	Measured, local
Transcribe	— (already local)	MLX Whisper	17–20 s per 7-min lecture
Translate	Gemini 2.5 Flash	**Gemma 4 31B** (MLX 4-bit, Apache-2.0)	59 s for the clip (12B via ollama: 25 s)
Hindi speech	Sarvam API	**Qwen3-TTS 1.7B** (MLX, Apache-2.0)	**1.6× real time**, cloned (first run: ~4×)

- The "Fully local" demo card was produced this way — **₹0 per lecture, nothing leaves**

Roadmap: smarter time alignment

We surveyed the automatic-dubbing literature (Amazon, Microsoft, IIT Madras, IWSLT) and re-based the alignment design on it — full notes in `docs/timing-alignment.md`:

Already implemented — word budgets in the translation prompt; asymmetric speed clamp; spill-into-pause instead of truncation; trailing-silence trimming.

Next:

1. **TTS-native pace instead of waveform stretching** — Sarvam exposes `pace` (0.3–3.0×); synthesis at the right speed beats resampling in every published comparison.
2. **Re-translate outliers once** — segments still needing > 1.3× speed get one "shorten by 25%" LLM pass; ~25% of En→Hi segments are expected to qualify.
3. **Sentence-boundary segmentation** with word-level timestamps (≈ +4.5 BLEU over blind packing).

Summary

- **Any lecture → natural Hindi dub in minutes**: one command, six automated stages, technical vocabulary preserved.
- **Measured, not estimated**: a 7-min lecture in ~3 minutes for under ₹10; ≈ ₹80–110 per lecture-hour; a 40-lecture course for the price of a textbook.
- **20–100× cheaper** than commercial dubbing platforms, with editorial control none of them offer.
- Open source (MIT): **github.com/nipunbatra/vidvaani**
- Live demos: **nipunbatra.github.io/vidvaani**

> Next: faculty-in-the-loop review, more Indian languages, and a student comprehension study.